

Constant-Term Borel Complex and Local Loebification

Autonomous discussion Pass 96 compares the Pass-95 constant-term Borel complex with the local Loeb sheafification from Pass 91.

For finite support S , the compact skeleton of the global constant-term complex is

$$C_B^{\text{int}}(S) = \mathbb{Q}^\times \ltimes \left[\mathbb{Z} \rightarrow \prod_{p \in S} \mathbb{Z}_p \right],$$

with the degree-0 map diagonal. The local Loebified target is

$$C_L(S) = (\mathbb{Q}^\times)^S \ltimes \left[\mathbb{Z}^S \rightarrow \prod_{p \in S} \mathbb{Z}_p \right],$$

with the degree-0 map coordinatewise.

The comparison is the map of two-term complexes

$$[\mathbb{Z} \rightarrow \prod_{p \in S} \mathbb{Z}_p] \rightarrow [\mathbb{Z}^S \rightarrow \prod_{p \in S} \mathbb{Z}_p],$$

diagonal in degree 0 and identity in degree 1. On unipotent H^1 it gives

$$0 \rightarrow \mathbb{Z}^S / \Delta \mathbb{Z} \rightarrow (\prod_{p \in S} \mathbb{Z}_p) / \Delta \mathbb{Z} \rightarrow \prod_{p \in S} (\mathbb{Z}_p / \mathbb{Z}) \rightarrow 0.$$

Therefore the precise unipotent kernel lost by local Loebification is

$$K_S = \mathbb{Z}^S / \Delta \mathbb{Z} \cong \mathbb{Z}^{|S|-1}.$$

At finite level N , the lost kernel has size $N^{|S|-1}$.

The Levi comparison is different. The diagonal map

$$\mathbb{Q}^\times \rightarrow (\mathbb{Q}^\times)^S$$

has trivial kernel. Local Loebification loses global Levi coherence as the quotient

$$(\mathbb{Q}^\times)^S / \Delta \mathbb{Q}^\times,$$

not as a second kernel.

Thus the best formulation is a map of two-term complexes plus stackification/local constant-term projection. Pure Hausdorff reflection captures only the unipotent quotient and misses the Levi decentralization.

The finite checker `artifacts/reports/pass96-constant-term-local-loebification-check.json` verifies the two-term complex map, the finite kernel sizes, singleton vanishing, multi-prime nontriviality, and the Levi quotient calculation.